Annotated Bibliography

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References

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[2] Madeleine Ernst, Kyo Bin Kang, Andrés Mauricio Caraballo-Rodríguez, Louis-Felix Nothias, Joe Wandy, Christopher Chen, Mingxun Wang, Simon Rogers, Marnix H. Medema, Pieter C. Dorrestein, and Justin J.J. van der Hooft. Molnetenhancer: Enhanced molecular networks by integrating metabolome mining and annotation tools. *Metabolites*, 9(7), 2019.

[3] Wenkai Fan. Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai–tibet plateau and high mountain regions. Undergraduate thesis, The College of Wooster, Wooster, OH, 2025. Unpublished thesis.

This unpublished work examines detection and quantification of organic pollutants in high-altitude environments, contributing to understanding contamination in sensitive ecosystems.

[4] Peter K.T. Newton-S.R. et al. Fisher, C.M. Approaches for assessing performance of high-resolution mass spectrometry–based non-targeted analysis methods. *Anal Bioanal Chem*, 414:6455–6471, 2022.

[5] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and Oliver Fiehn. Identification of small molecules using accurate mass ms/ms search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

This review explains how to identify small molecules using high-resolution MS/MS techniques. It highlights the importance of spectral libraries that include experimental data and computer-generated spectra. The article covers methods for creating in silico spectra using various software tools and developing new databases. It also stresses the need for proper data curation

and quality control. From my point of view, it provides valuable insights into how computational approaches and shared spectral resources can aid in analyzing complex samples. These methods are especially helpful for identifying unknown compounds in environmental or natural product research.

[6] Feerick A.-Braun J. M. Feng Y. L. Hall A. Koelmel J. Manzano C. Newton S. R. Pennell K. D. Place B. J. Godri Pollitt K. J. Prasse C. Young J. A. Manz, K. E. Non-targeted analysis (nta) and suspect screening analysis (ssa): A review of examining the chemical exposome. *Journal of Exposure Science Environmental Epidemiology*, 33(4):524–536, 2023.

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[10] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

This viewpoint article is essential for non-targeted analysis (NTA) research using HR-MS, such as the computational identification of unknown contaminants in remote environments. It proposes a five-level confidence system for molecular identification, addressing the limitations inherent when reference standards are absent. The framework provides the necessary rigor for reporting the results of computational tools like molecular networking or cluster analysis. Specifically, Level 3 (Tentative candidate(s)) helps classify structures where computational methods suggest isomers, while Levels 4 and 5 establish clear reporting standards for masses and formulas that serve as critical input or nodes in network-based analysis, enabling transparent data exchange and future traceability of unknown molecules.

[11] Weiting Ye, Jingcheng Li, and Xianfa Cai. Mfgnn: Multi-scale feature-attentive graph neural networks for molecular property prediction. *Journal of Computational Chemistry*, 46(3):e70011, 2025.

This study introduces MfGNN, a multi-scale feature-attentive graph neural network designed for molecular property prediction by capturing both atom-level and fragment-level structural features. While its primary focus is on

predicting molecular properties, the methodologies—particularly the integration of fragment-based representations and inter-fragment relationship modeling—offer valuable insights for non-targeted analysis in high-resolution mass spectrometry. Its ability to systematically incorporate substructural information and learn complex molecular relationships aligns with the objectives of molecular networking and cluster analysis, moving beyond traditional structure assignment based solely on mass spectra. Incorporating such graph-based machine learning frameworks can enhance the annotation, classification, and clustering of unknown molecules in environmental samples, especially in remote and contaminated regions. Therefore, this work provides a computationally advanced approach that can significantly contribute to the development of chemoinformatics tools for high-resolution mass spectrometry in non-targeted environmental studies.